INCH-POUND

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5 March 2026
SUPERSEDING
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10 June 1992

DETAIL SPECIFICATION
CLOTH, TROPICAL: POLYESTER AND WOOL

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for a polyester and wool tropical cloth, washable, in top dye and piece dye fabrics.

1.2 Classification. The cloth is available in the following types, as specified (see 6.2).

1.2.1 Types.

Type I - DELETED

Type II – DELETED

Type III – Top dye, 75/25 polyester/wool, khaki shade M-1

2. APPLICABLE PUBLICATIONS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-655 - Provisions for Evaluating Quality of Cloth, Wool, Worsted and Wool Blends

(Copies of these documents are available online at https://quicksearch.dla.mil)

Comments, suggestions, or questions on this document should be sent to trpsptctstandardiz@dla.mil. Since contact information can change, verify the currency of the address information using Acquisition Streamlining and Standardization Information System ASSIST online database https://assist.dla.mil.

AMSC: N/A FSC: 8305

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2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation or contract.

UNITED STATES DEPARTMENT OF AGRICULTURE

Wool Grades and Standards

(Copies of these documents are available online at https://www.ams.usda.gov/)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN ASSOCIATION OF TEXTILE CHEMISTS AND COLORISTS (AATCC)

AATCC EP9 - Visual Assessment of Color Difference of Textiles
AATCC TM8 - Test Method for Colorfastness to Crocking: Crockmeter
AATCC TM15 - Test Method for Colorfastness to Perspiration
AATCC TM16.3 - Test Method for Colorfastness to Light: Xenon-Arc
AATCC TM20A - Test Method for Fiber Analysis: Quantitative
AATCC TM61 - Test Method for Colorfastness to Laundering: Accelerated
AATCC TM81 - Test Method for pH of the Water-Extract from Wet Processed Textiles
AATCC TM132 - Test Method for Colorfastness to Drycleaning
AATCC TM158 - Test Method for Dimensional Changes on Drycleaning in Perchloroethylene: Machine Method

(Copies of these documents are available online at www.aatcc.org.)

AMERICAN SOCIETY FOR QUALITY (ASQ)

ASQ/ANSI Z1.4 - Sampling Procedures and Tables for Inspection by Attributes

(Copies of this document are available online at https://asq.org.)

ASTM INTERNATIONAL

ASTM D1777 - Standard Test Method for Thickness of Textile Materials
ASTM D2165 - Standard Test Method for pH of Aqueous Extracts of Wool and Similar Animal Fibers
ASTM D3512/3512M - Standard Test Method for Pilling Resistance and Other Related Surface Changes of Textile Fabrics: Random Tumble Pilling Tester
ASTM D3775 - Standard Test Method for End (Warp) and Pick (Filling) Count of Woven Fabrics
ASTM D3776/D3776M - Standard Test Methods for Mass Per Unit Area (Weight) of Fabric
ASTM D5034 - Standard Test Method for Breaking Strength and Elongation of Textile Fabrics (Grab Test)

(Copies of these documents are available online at www.astm.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 Inspections.

3.1.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.3.

3.1.2 Conformance inspection. When specified (see 6.2), a sample shall be subjected to conformance inspection in accordance with 4.3.

3.1.3 Standard sample. The finished cloth shall match the standard sample for shade and shall be equal to or better than the standard sample with respect to all characteristics for which the standard sample is referenced (see 6.3)

3.3 Material properties.

3.3.1 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous lifecycle costs.

3.3.2 Fiber.

3.3.2.1 Wool. The stock shall be wool fleece, pulled wool, or both. Wool fiber grade shall be no lower than 62's US when tested as specified in 4.4.1.1, as specified by the U.S. Department of Agriculture Grades and Standards (see 2.2). Wool fibers shall be between 3 inches and 3-1/2 inches in length, and shall be spun on the worsted spinning system when blended with the appropriate polyester fiber (see 3.3.2.2).

3.3.2.2 Polyester. The polyester stock shall be semi-dull polyester fiber made from polyethylene glycol terephthalate, either homopolymer or modified polymer, as appropriate. The fiber shall be 2.5 to 3.0 denier and the minimum average fiber length shall be three inches when tested as specified in 4.4.1.2.

3.3.3 Polyester/wool yarn. The polyester and wool fibers shall be blended and spun on the worsted system for both warp and filling yarns. The yarns shall be two-ply yarns in both warp and filling. The finished cloth shall contain the following fiber percentages (±3 percent), based on the oven-dry weight, when tested as specified in 4.4.1.3. Top dyed cloth shall be a 75 percent polyester / 25 percent wool blend, and piece dyed cloth shall be a 80 percent polyester / 20 percent wool blend.

3.3.4 Color.

3.3.4.1 Shade. The shade shall be shade M-1 for type III cloth. The finished cloth shall match the standard sample (see 6.3) when tested under artificial daylight (D65) with a color temperature of 6500 ±200 Kelvin (K) and incandescent A at a temperature of 2856 ±200 K. Type III shade shall be obtained through top dyeing. The shade of the selvage shall approximate the shade of the cloth.

3.3.4.2 Colorfastness. When tested as specified in 4.5.1, colorfastness to laundering (after 4 cycles), wet drycleaning, perspiration (acid), light and crocking shall be assessed. The finished cloth shall show colorfastness equal to or better than the standard for polyester and wool. When no standard is available, colorfastness shall be a minimum rating of 4 for all types.

3.4 Finished cloth.

3.4.1 Physical requirements. When tested as specified in 4.5.1, the finished cloth shall conform to the requirements specified in table I.

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TABLE I. Physical requirements.

	Type III
Yarns per inch (±1)	
Warp	55
Filling	47
Breaking Strength, pounds (minimum)	
Warp	100
Filling	80
Shrinkage after laundering, percent (maximum)	
Warp	2.5
Filling	2.5
Weight (ounces per square yard)	5.2 to 5.7
Weave	Plain (1 by 1)
pH	5.5 – 8.5
Pilling, rating (minimum)	4

3.4.2 Length, width and put up. Unless otherwise specified (see 6.2), the cloth shall be furnished on rolls in continuous lengths, each not less than 50 yards, and in full widths not less than 60 inches, not including selvage. Selvages shall be no more than 65 percent greater in thickness than the body of the cloth. The finished cloth shall lay flat without puckers, skew or creases when laid out for cutting.

3.4.3 Finish. The cloth shall be scoured, heat set, sheared and decated so as to provide a finish equal to that of the standard sample (see 6.3) for the applicable type.

3.4.4 Pilling. DELETED

3.4.5 pH. DELETED

3.5 Marking.

3.5.1 Back marking. The back of the cloth shall be marked “BACK” at both ends of the roll with letters not less than 1/2 inch in height. The marking shall be clearly legible in any indelible marking used commercially. Markings shall be examined as specified in 4.5.3.3.

3.5.2 Wool content labeling. Cloth manufactured under this specification shall be labeled in accordance with the Wool Products Labeling Act of 1939. Markings shall be examined as specified in 4.5.3.3.

3.6 Shelf life. For fabrics sold to the government, shelf life requirements will be specified in the contract or order.

3.7 Workmanship. The finished cloth shall conform to the quality of product established by this document. Defects shall not exceed the established acceptance criteria (see 4.5.3 and 6.2) and shall not adversely affect the serviceability, appearance, and uniformity of the product.

4. VERIFICATION

4.1 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).

b. Conformance inspection (see 4.4).

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4.2 Inspection conditions. Unless otherwise specified (see 6.2), all inspections shall be performed in accordance with the test conditions specified in the referenced test methods and 4.4.1 and 4.5.2.

4.3 First article inspection. The first article, submitted in accordance with the contract or order (see 6.2), shall be examined in accordance with 4.4.1 through 4.4.2 for compliance with the design, configuration, workmanship, dimensional requirements, and testing as specified in this document.

4.4 Conformance inspection. Conformance samples, submitted in accordance with the contract or order (see 6.2), shall be examined in accordance with 4.4.1 through 4.4.2 for compliance with the design, configuration, workmanship, dimensional requirements, and testing as specified in this document.

4.4.1 Fiber.

4.4.1.1 Wool grading. Wool shall be examined for grade, using the micron system, visually and through standard laboratory testing practices used in the wool industry.

4.4.1.2 Polyester fiber. The polyester fiber shall meet the requirements of 3.3.2.2 visually and through standard laboratory testing practices used in the fiber industry.

4.4.1.3 Yarns. The blended yarns shall meet the requirements of 3.3.3 when tested visually and in accordance with AATCC TM20A.

4.4.2 Finished cloth sampling and acceptance criteria. Unless otherwise specified in the contract (see 6.2), sampling for inspection (see 4.5.1) shall be performed in accordance with table II and ASQ/ANSI Z1.4. Sampling for examination specified in 4.5.3 shall be in accordance with 4.5.3. The sample unit shall be one yard of cloth, and the lot size shall be expressed in yards of finished cloth. The presence of excessive defects or failure of any testing as specified (see 6.2) shall be cause for rejection of the lot.

TABLE II. Material sampling.

Lot size	Sample size
800 or less	2
801 through 22,000	3
22,001 and over	5

4.4.2.1 Certificates of compliance. Where certificates of compliance are submitted, the government reserves the right to test such items to verify the validity of the certification.

4.5 Inspection methods.

4.5.1 Material examinations and tests. In accordance with 4.1, components and materials shall be examined and tested in accordance with the examinations and tests specified in table III, unless otherwise excluded, amended, or modified in this document or applicable procurement documents (see 6.2).

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TABLE III. Finished cloth verification.

Characteristic	Requirement	Verification
Shade	3.3.4.1	AATCC EP9/1
Yarns per inch	Table I	ASTM D3775
Weight	Table I	ASTM D3776/D3776M, Option C
Breaking strength	Table I	ASTM D5034
Colorfastness to:	3.3.4.2	
Perspiration		AATCC TM15
Light		AATCC TM16.3, Option 3
Laundering (after 4 cycles)		AATCC TM61, Option 1A
Wet dry cleaning		AATCC TM132 1/
Crocking		AATCC TM8
Shrinkage LaunderingDELETED	Table I	AATCC TM135
Weave	Table I	Visual
pH	3.4.5	AATCC TM81
Pilling	Table I	ASTM D3512/D3512M
Selvage thickness	3.4.2	ASTM D1777
Overall length and width	3.4.2	Visual

1/ Results recorded as pass or fail.

2/ AATCC 16.3 shall use 40 AATCC Fading Units.

3/ Each shrinkage test specimen shall be visually examined for creping and the results reported as "pass" or "fail". Failure of any one of the test specimens shall be cause for rejection.

4.5.2 In-process examination. Visual and dimensional examinations shall be made at any point or during any phase of the manufacturing process to determine whether construction details, which cannot be examined in the finished product, are in accordance with the requirements specified in section 3. Materials which contain defects and damages as defined in the visual examination of 4.5.3 shall be removed from production. Any in-process nonconformance remaining in the finished cloth shall be classified as a defect in accordance with the visual examination of 4.4.

4.5.3 Visual examination. The cloth shall be examined for defects in color, fiber, construction, workmanship, and marking in accordance with MIL-STD-655. If the acceptance criteria differ from those defined in MIL-STD-655, they shall be specified in the contract or order (see 6.2).

4.5.3.1 Yard-by-yard examination. Each roll in the sample shall be examined for defects on the face side only. When the yardage on the roll does not exceed 100 yards, the entire roll shall be inspected. When the yardage on the roll exceeds 100 yards, only 100 yards shall be examined.

4.5.3.2 Examination for length. Total yardage on the roll shall be withing two yards of the total length marked on the label. The total yardage in the lot shall be within one percent of the total reported for the lot size.

4.5.3.3 Examination for labeling and marking. Each roll in the lot shall be labeled in such a manner that compliance with the wool Products Labeling Act can be determined until the roll is fully expended. Back side markings, in accordance with 3.5.1, shall be verified as part of the visual examination.

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The cloth is intended for use in production of service and dress shirts worn by personnel of the United States armed Services.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Type(s) required (see 1.2).
c. When first article inspection is required (see 3.1.1 and 4.3).
d. When conformance inspection is required (see 3.1.2 and 4.4).
e. Shelf life requirements (see 3.6).
f. First article and conformance inspection sampling and acceptance criteria (see 4.3 and 4.4).
g. Exceptions to inspection conditions (see 4.2).
h. Exceptions to or additions to component and material examination and tests (see 4.5.1).
i. Visual examination inspection conditions, acceptance criteria and AQLs (see 3.7 and 4.5.3).
j. Packaging required (see 5.1).

6.3 Information requests. For access to information such as specification and standard shade samples of cloth, go to https://www.dla.mil/TroopSupport/ClothingandTextiles/SpecRequest.aspx for the Defense Logistics Agency Troop Support's Clothing and Textile Specification/Pattern/Drawing Request form. Complete the request form and then submit the form. Requests to use equivalent materials or components or to make changes to the cloth should be sent to the contracting officer for approval by the military services.

6.7 Certificate of compliance. Certificates of compliance may be accepted by the contracting agency in lieu of test data. It is recommended that vendors contact the contracting officer prior to using certificates of compliance to address any test requirement. The government reserves the right to verify the accuracy of any data on certificates of compliance.

6.8 Shelf-life. This specification covers items where the assignment of a Federal shelf-life code is a consideration. Specific shelf-life requirements should be specified in the contract or purchase order, and should include, as a minimum, shelf-life code, shelf-life package markings in accordance with MIL-STD-129 or FED-STD-123, preparation of a materiel quality storage standard for type II (extendible) shelf-life items, and a minimum of 85 percent shelf-life remaining at time of receipt by the Government. These and other requirements, if necessary, are in DoDM 4140.27 Volume 1, *DoD Shelf-Life Management Program: Program Administration,* and DoDM 4140.27 Volume 2*, DoD Shelf-Life Management Program: Materiel Quality Control Storage Standards*. The shelf-life codes are in the Federal Logistics Information System Total Item Record. Additive information for shelf-life management may be obtained from DoDM 4140.27 Volumes 1 and 2, or the designated shelf-life Points of Contact (POC). The POC should be contacted in the following order: (1) the Inventory Control Points that manage the item and (2) the DoD Service and Agency administrators for the DoD Shelf-Life Program. Appropriate POCs for the DoD Shelf-Life Program can be contacted through the DoD Shelf-Life Management website: https://www.shelflife.dla.mil/.

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6.9 Subject term (key word) listing.

Khaki

Piece dye

Shirt

Top dye

6.10 Changes from previous issue. Marginal notations are used in this revision to identify changes with respect to the previous issue.

CONCLUDING MATERIAL

Custodian

Navy - MC

Preparing activity:

DLA-CT

(Project 8305-2025-038)

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